Automata Theory

Mathematical Notions & Terminology

CS32022

By Thisara Watawana

08 July 2026

1. Mathematical Notions & Terminology

Sets

- A set is a collection of objects treated as a single unit. The objects inside are called elements or members.
  - Examples: {56, 87, 1, 8}, {true, false}, {x, y, r, p}
- The order of describing a set doesn't matter, nor does repetition of its members.
  - Examples: {65, 12, 4} as same as {4, 12, 65} or {65, 12, 12, 4}
- Multisets: Track element occurrences/repetition.
  - Examples: {7} and {7, 7} are identical as sets, but distinct as multisets

1. Mathematical Notions & Terminology

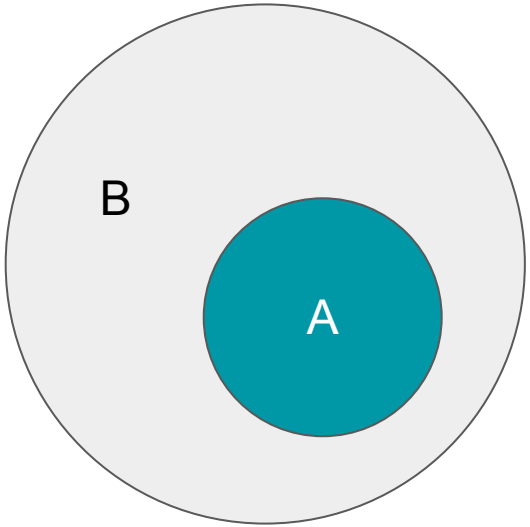
Sets

- Membership: $e \in S$ means e is an element of S.
- Non-membership: $e \notin S$ means e is not an element of S.
- Infinite set contains infinitely many elements.
  - Example: The set of integers $\mathbb{Z}$ is written as {..., −2, −1, 0, 1, 2, ...}.
- The set with zero members is called the empty set and is written $\varnothing$.
- Set-builder/Rule notation: set containing elements according to some rule
  - Example: B = {n | n is an even integer}

1. Mathematical Notions & Terminology

Subsets

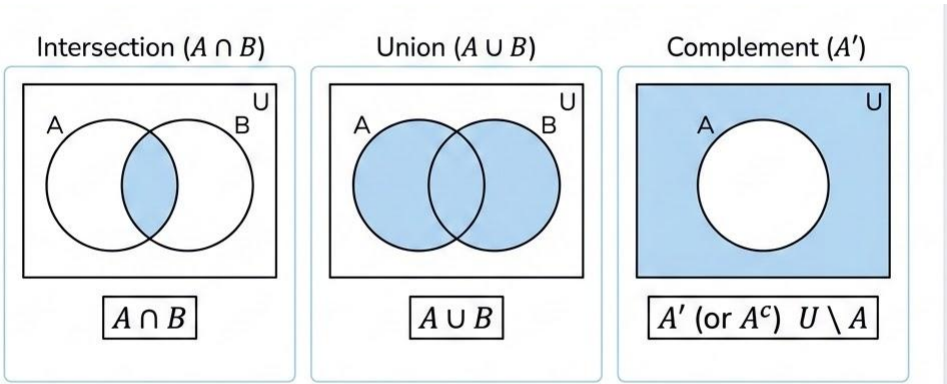
- Subset ($A \subseteq B$): Every element of A is also an element of B.
- Proper Subset ($A \subsetneq B$): A is a subset of B, but A is not equal to B (there is at least one element in B that isn't in A).



1. Mathematical Notions & Terminology

Set Operations

- Union (A ∪ B): Everything that is in A, OR in B, or in both.
- Intersection (A ∩ B): Elements that live in both A AND B simultaneously.
- Complement (A): Everything in the universal domain that is not in A.



1. Mathematical Notions & Terminology

Sequences and Tuples

- Sequence: A list of objects where order and repetition matter.
  - Written inside parentheses.
  - For example, (7, 21, 57) is not the same as (57, 7, 21).
  - Duplicates are respected: (7, 21, 57, 7) is a valid sequence of length 4.
- Tuples: A finite sequence of length k is called a k-tuple.
  - A 2-tuple is specifically called an ordered pair.
  - Example: (q_0, a) is an ordered pair commonly used to trigger a state transition.

1. Mathematical Notions & Terminology

The Power Set

- The Power Set of a set S, denoted as $\mathcal{P}(S)$ or 2^S, is the set of all possible subsets of S.
- The Empty Set rule: The empty set $\varnothing$ is always a member of any power set.
- If a finite set S has n elements, its power set $\mathcal{P}(S)$ will always contain exactly 2^n elements.
- Example: If $S = \{x, y\}$, then $\mathcal{P}(S) = \{\varnothing, \{x\}, \{y\}, \{x, y\}\}$.

1. Mathematical Notions & Terminology

The Cartesian Product

- The Cartesian Product of two sets A and B, written as $A \times B$, is the set of all ordered pairs where the first element belongs to A and the second belongs to B.
- Formal Rule: $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$
- Scaling up (k-sets): We can cross multiply multiple sets ($A_1 \times A_2 \times \ldots \times A_k$) to construct a set of k-tuples.

1. Mathematical Notions & Terminology

Exercise 01

- Imagine you are programming the controller for an automated highway toll gate. The gate starts out with the barrier arm down to prevent cars from passing through without paying.
- When a driver taps their payment card, the barrier arm lifts. Once the vehicle clears the safety sensor on the other side, the arm immediately lowers again to block the next car.
- Compute the Cartesian Product of states $State \times Input$

1. Mathematical Notions & Terminology

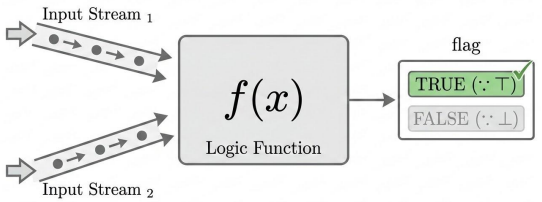
Functions

- A function is an object that sets up an input-output relationship. A function f takes an input x and produces an output y, written as $f(x) = y$.
- The set of all possible inputs to a function is its domain. The set of all possible allowable outputs is its codomain.
- We write $f\colon D \to R$ to state that f is a function with domain D and codomain R.

1. Mathematical Notions & Terminology

Functions

- Onto Functions: A function is onto if it hits every single element in its codomain (meaning its actual range equals its codomain).
- Multi-Input Domain: If a function takes multiple inputs, its domain is expressed as a Cartesian product.
    - Example: $f\colon A \times B \to C$ takes an ordered pair (a, b) as a single input.
- Predicates: A predicate is a specific type of function whose codomain is strictly the boolean set {True, False} (or {0, 1}).



1. Mathematical Notions & Terminology

Binary Relations & Equivalence

- Binary Relation: A predicate whose domain is the Cartesian product of two sets ($A \times A$).
- It says whether a specific relationship holds true between two elements. We often write it inline: a R b.
- Equivalence Relation: A relation is an equivalence relation if it satisfies three strict criteria:
    - Reflexive: a R a is always true for every element.
    - Symmetric: If a R b is true, then b R a must be true.
    - Transitive: If a R b and b R s are true, then a R s must be true.
    - Example: The relation "congruence modulo 7" (written $a \equiv b \pmod 7$) is a classic equivalence relation.

1. Mathematical Notions & Terminology

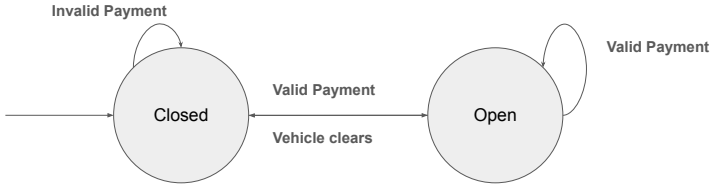
Exercise 02

- Consider the set of integers $\mathbb{Z}$, and consider the relation $a \equiv b \pmod 3$ (meaning a and b leave the exact same remainder when divided by 3). Take 60 seconds with your neighbor and group these numbers into their matching buckets: {−3, −2, −1, 0, 1, 2, 3, 4, 5, 6}. How many distinct buckets do you get?

1. Mathematical Notions & Terminology

Graph Theory

- A graph G is an object consisting of two sets: a set of vertices/nodes (V) and a set of edges (E). We write it as $G = (V, E)$.
- Undirected Graphs: The edges have no directional orientation. An edge is simply an unordered set of two nodes: $\{u, v\}$.
- The Degree of a Node: The number of edges attached to a specific node.



1. Mathematical Notions & Terminology

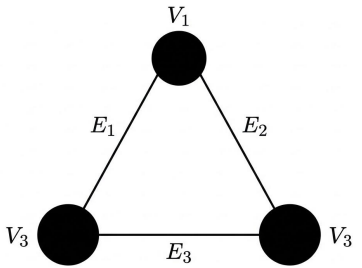
Directed Graphs & Self-Loops

- Directed Graphs (Digraphs): Edges have arrows indicating a specific direction from a source node to a target node.
- An edge in a directed graph is an ordered pair (u, v) rather than a set, meaning the edge starts at u and goes to v.
- In-Degree vs. Out-Degree:
  - In-Degree: The number of arrows entering a node.
  - Out-Degree: The number of arrows leaving a node.
- Self-Loops: An edge can start and end at the exact same node, written as (u, u).

1. Mathematical Notions & Terminology

Paths, Cycles, and Trees

- **Path**: A sequence of nodes connected by edges. A simple path is a path that never repeats the same node twice.
- **Cycle**: A path that starts and ends at the exact same node. A simple cycle contains no repeated nodes except for the matching start and end node.
- **Connected Graph**: An undirected graph where a path exists between every single pair of nodes.
- **Tree**: A connected graph that contains no simple cycles. A tree always looks like a branching hierarchy.



1. Mathematical Notions & Terminology

Exercise 03

- Draw the directed graph $G = (V, E)$ defined by these two sets:
  - $V = \{q_0, q_1, q_2\}$
  - $E = \{(q_0, q_1), (q_1, q_2), (q_2, q_2), (q_2, q_0)\}$

1. Mathematical Notions & Terminology

Strings and Languages

- **Alphabet (Σ)**: Any finite, non-empty set of symbols.
  - Binary Alphabet: Σ = {0, 1}
  - English Alphabet: Σ = {a, b, c, …, z}
- **String**: A finite sequence of symbols from a given alphabet, written next to each other.
  - Example: If Σ = {0, 1}, then 01101 is a string over Σ.
- **String Length (|w|)**: The number of symbols contained in a string w.
  - If w = 01101, then |w| = 5.

1. Mathematical Notions & Terminology

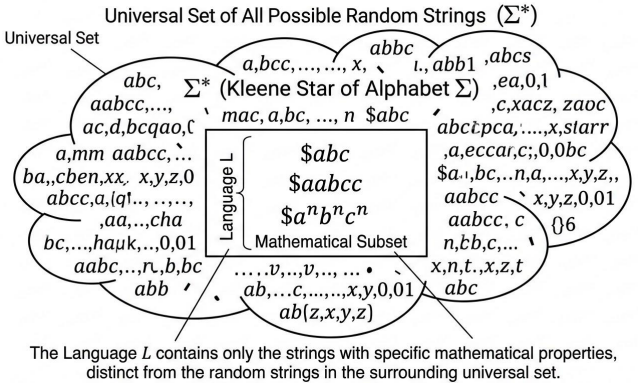
The Empty String & Substrings

- **The Empty String (ε)**: The unique string containing exactly zero symbols. Its length is always zero:
  - $|\varepsilon| = 0$.
- **Substrings**: A string z is a substring of w if z appears consecutively anywhere inside w.
  - Example: If w = cadillac, then dill is a substring of w.
- **String Concatenation**: Combining two strings x and y by pasting y onto the end of x to form a single joint string, written as xy.
- **Multi-concatenation power identity**: x^k means repeating the string x exactly k times. (Note: $x^0 = \varepsilon$).

1. Mathematical Notions & Terminology

Formal "Language"

- A Language is simply a set of strings over a chosen alphabet.
- The Kleene Star (Σ*): The infinite set of all possible finite strings that can ever be built using the symbols from alphabet Σ.
- Mathematical Truth: A language *L* is a subset of the universal string space ($L \subseteq \Sigma^*$).



1. Mathematical Notions & Terminology

Ordering Strings

- **Lexicographic Ordering**: The standard dictionary order where strings are sorted alphabetically based on the underlying symbol order.
    - Short strings can come after long strings if they start with a later letter.
- **Shortlex/String Ordering**: The preferred ordering method in automata theory. Strings are sorted strictly by length first, and then ties are broken alphabetically.
    - Example over alphabet {0, 1} in Shortlex order:

```
ε, 0, 1, 00, 01, 10, 11, 000, 001, …
```

1. Mathematical Notions & Terminology

Exercise 04

Let our alphabet be $\Sigma = \{0, 1\}$. I have defined a formal language as follows:

$$L = \{w \in \Sigma^* \mid w \text{ starts with 0 and has an even length}\}$$

Write down the first 4 shortest strings that belong to this language L, sorted in proper Shortlex order.

1. Mathematical Notions & Terminology

Boolean Logic

- Boolean Logic: A mathematical system built on two core values: TRUE (1) and FALSE (0).
- Core Operations:
  - Negation / NOT (¬ or $\bar{p}$): Flips the truth value.
  - Conjunction / AND (∧): True only if both inputs are true.
  - Disjunction / OR (∨): True if at least one input is true.

1. Mathematical Notions & Terminology

Boolean Logic

- Advanced Structural Operators:
  - Exclusive OR / XOR (⊕): True if the inputs are different (one is true, one is false). False if they are identical.
  - Equality / Equivalence (↔ or ≡): True if the inputs are identical (both true or both false). This is the direct logical opposite of XOR.
  - The Conditional / Implication (→): p → q means "If p, then q". It is only false when p is true but q is false.

1. Mathematical Notions & Terminology

The Distributive Laws

- Describes how AND and OR operations interact when nested.
- Law 1: AND distributes over OR
  - $a \land (b \lor c) = (a \land b) \lor (a \land c)$
- Law 2: OR distributes over AND
  - $a \lor (b \land c) = (a \lor b) \land (a \lor c)$

1. Mathematical Notions & Terminology

Exercise 05

- Imagine you are reviewing a code for a critical state-machine safety check. There is a massive conditional statement to evaluate three boolean flags: *A* (System Armed), *B* (Sensor High), and *C* (Timer Active).
- The code reads:
  - `if (A and B) or (A and C) or (A and (not B or not C)):`
- This translates to the formal boolean expression:
  - $(A \land B) \lor (A \land C) \lor (A \land (\neg B \lor \neg C))$
- Use the Distributive Laws we just learned to refactor and radically simplify this expression. What is the absolute shortest equivalent logic?

1. Mathematical Notions & Terminology

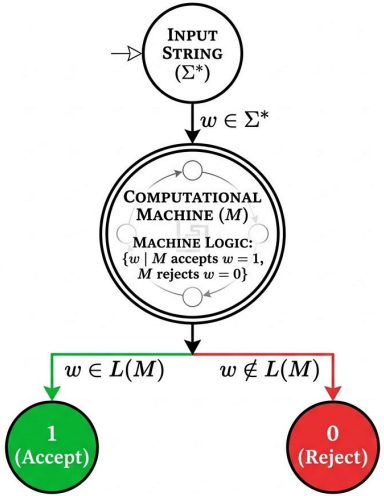
Transitions as Logic

- Under the hood, an automaton's control unit is just a collection of physical logic gates (AND, OR, NOT).
- Moving from one state to another is a conditional branch evaluated using boolean math.
- If a machine is in State q_0 AND reads the symbol 1, the transition evaluates to TRUE and the machine jumps to the next state.
- Complex routing inside a machine is just a network of these binary decisions happening simultaneously.

1. Mathematical Notions & Terminology

Problems as Boolean Predicates

- A computer's job in this course is to look at a string w and decide if it is valid or invalid.
- We model this computation as a Predicate Function (a funct that outputs a boolean value).
- Mathematical Mapping:
    - $f(w)$ {1, 0} Output 1 (True): The string belongs to the language ACCEPT.
    - Output 0 (False): The string violates the rules REJECT.



1. Mathematical Notions & Terminology

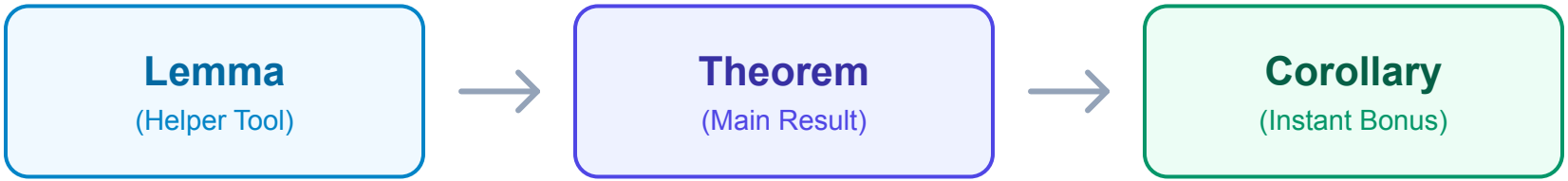
Definitions, Theorems, & Proofs

- **Definition**: An explicit, unambiguous specification of a mathematical object or rule (e.g., $G = (V, E)$).
- **Theorem:** A mathematical statement or property claimed to be 100% true under given definitions.
- **Proof:** A rigorous, step-by-step logical deduction demonstrating that a theorem holds true without exception.

1. Mathematical Notions & Terminology

Lemmas & Corollaries

- **Lemma:** A smaller helper theorem proven before a main theorem to break complex logic into modular, readable parts.
- **Corollary:** A bonus truth that follows naturally and immediately after a main theorem has been proven.



1. Mathematical Notions & Terminology

Why Proofs & Definitions Matter in Automata

- **Infinite Input Spaces:** Software test suites can only check finite inputs. A proof guarantees a machine works across infinitely many strings (Σ^*).
- **Proving Impossibility:** To prove a machine cannot solve a problem (e.g., validating nested HTML tags), we must use a formal proof as testing cannot prove a negative.
- **Algorithm Correctness:** We use proofs to guarantee that machine transformations (e.g., converting NFAs to DFAs) preserve identical behavior.

1. Mathematical Notions & Terminology

Exercise 06

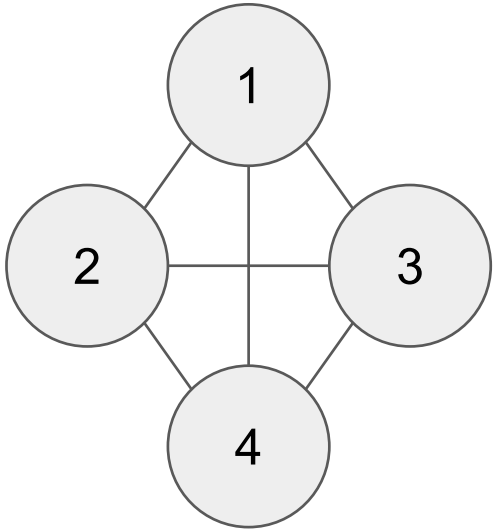
Match each statement below to its correct role: [Definition | Lemma | Theorem | Proof | Corollary]

- **A**: "An Even Group is any group whose total count N can be written as $N = 2k$ for some integer k."
- **B**: "Algebraic helper rule: $2a + 2b = 2(a + b)$."
- **C**: "Merging any two separate Even Groups always creates a group that can be split into 2 equal teams."
- **D**: "Let Group 1 have $2a$ players and Group 2 have $2b$ players. Total = $2a + 2b = 2(a + b)$. Since $(a + b)$ is an integer, the group satisfies Card A."
- **E**: "Merging four separate Even Groups can also always be split into 2 equal teams."

1. Mathematical Notions & Terminology

Proof Method 1: Proof by Construction

- **Theorem:** For every even integer $n \geq 4$, there exists a 3-regular graph (a graph where every node has a degree of exactly 3) containing n vertices.
- Do not argue abstractly. Give an algorithm to build the graph for any even n.
- In a proof by construction, your goal is to act like a software architect. When a theorem claims an object exists, you provide an algorithm that manufactures it.

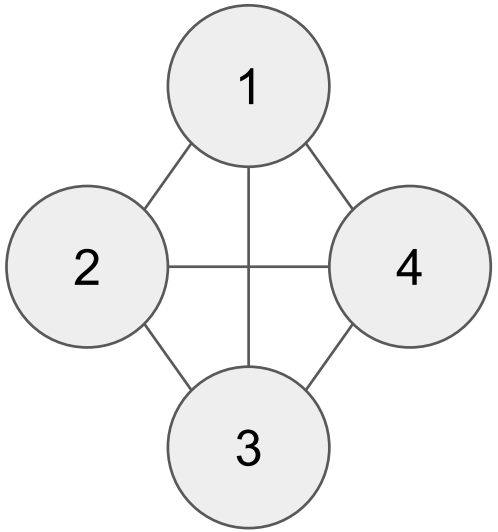


1. Mathematical Notions & Terminology

Proof Method 1: Proof by Construction

Every node has 2 outer ring edges + 1 cross spoke edge = 3 edges total. The graph exists for all even *n* 4.

$$E = \{ \{i, i+1\} \mid 1 \le i \le n-1 \} \cup \{ \{n, 1\} \} \cup \{ \{i, i + n/2\} \mid 1 \le i \le n/2 \}$$



1. Mathematical Notions & Terminology

Proof Method 2: Proof by Contradiction

- Theorem: $\sqrt{2}$ is an irrational number (it cannot be expressed as a ratio of two integers).
- We assume that the theorem is false and then show that this assumption leads to an obviously false consequence, called a contradiction.
- Assume the opposite is true: $\sqrt{2} = ab$ (where the fraction is reduced to its lowest terms).

1. Mathematical Notions & Terminology

Proof Method 2: Proof by Contradiction

- **Assume Opposite:** Assume $\sqrt{2}$ is rational. Therefore, $\sqrt{2} = a/b$, where a/b is a fraction reduced to lowest terms (meaning a and b share no common factors).
- Square both sides: $2 = a^2/b^2 \Rightarrow a^2 = 2b^2$.
- Since $a^2 = 2b^2$, a^2 must be even, which means a must be even ($a = 2k$).
- $(2k)^2 = 2b^2 \Rightarrow 4k^2 = 2b^2 \Rightarrow 2k^2 = b^2$.
- Since $b^2 = 2k^2$, b^2 is even, which means b must also be even!
- **The Contradiction:** If both a and b are even, the fraction a/b was not reduced to lowest terms. This contradicts our starting assumption. Therefore, $\sqrt{2}$ must be irrational.

1. Mathematical Notions & Terminology

Proof Method 3: Proof by Induction

- **Theorem:** For any finite set S containing n elements, the power set $\mathscr{P}(S)$ contains exactly 2n elements.
- Every proof by induction consists of two parts, the **basis** and the **induction step**. Each part is an individual proof on its own. The basis proves that $P(1)$ is true. The induction step proves that for each $i \geq 1$, if $P(i)$ is true, then so is $P(i + 1)$.

1. Mathematical Notions & Terminology

Proof Method 3: Proof by Induction

- **Step 1: Base Case ($n = 0$)**:
  - A set with 0 elements is the empty set $S = \varnothing$.
  - Its power set is $\mathscr{P}(\varnothing) = \{\varnothing\}$, which has 1 element.
  - Since $2^0 = 1$, the base case holds true!

1. Mathematical Notions & Terminology

Proof Method 3: Proof by Induction

- **Step 2: Inductive Step**:
  - Assume a set with k elements has 2^k subsets.
  - Prove for $k + 1$ elements: Add one new element x to the set. Every existing subset stays as it was (there are 2^k of these).
  - Every existing subset can also be cloned by adding element x into it (another 2^k subsets).
  - Total subsets = $2^k + 2^k = 2 \cdot 2^k = \mathbf{2^{k+1}}$.
- **Conclusion**: By induction, $|\mathscr{P}(S)| = 2^n$ for all $n \geq 0$.

Questions?

thisaraw@boffin.lk